

Day 14 Plan

Maths: Algebra + Consecutive Numbers · Reasoning: Mixed Puzzles · Science:
Biology 2: Food Production, Agriculture, Nutrition & Digestion

Today's schedule (about 5½ hours with breaks)

#	Block	Time	What you do	Where
0	Day 13 recall	15 min	10 questions from Day 13, no notes. Redo yesterday's mistakes.	Page 1-2
1	Maths	75 min	5 simplification questions for speed → formulas & methods (10 min) → 2-3 book questions per pattern, timed → 10 Testbook PYQs on this topic → log mistakes	Section A · book · Testbook
-	Break	10 min	Away from the screen	-
2	Reasoning	45 min	Learn the method (10 min) → 15 timed questions, Testbook PYQs first → log	Section B · Testbook
-	Break	10 min	Away from the screen	-
3	Science	110 min	8 lessons with quick checks (≈65 min) → 20-question practice set → Testbook PYQs: food production & agriculture; vitamins (S tier)	Section C · Testbook
-	Break	10 min	Away from the screen	-
4	General awareness	45 min	20 current-affairs MCQs (20 min) → static GA: Indian Railways facts · organisations & their HQs — Testbook GA PYQs on it (25 min)	CA PDF · Testbook
5	Close	15 min	Redo today's wrong questions · write 3 rules for tomorrow · send me your scores	Mistake log

Short day? Cut in this order: Testbook science PYQs → reasoning down to 10 questions → static GA down to 10 min. Never cut recall, maths or the science lessons. · Exam pace (average 54 sec a question): GA ~30 sec · Science ~45 · Reasoning ~50 · Maths ~65.

Block 0 — Day 13 recall (answers on the last page)

#	Question	Your answer
1	Who coined the word 'cell'?	
2	Powerhouse of the cell?	
3	'Suicide bags' of the cell?	
4	Number of chromosomes in a human body cell?	
5	Tissue that carries water in plants?	
6	A tendon connects?	
7	Is cardiac muscle voluntary?	
8	Who proposed the five kingdoms?	
9	Volume of a sphere of radius 21?	
10	Circle facing the centre: left = ?	

Fewer than 8 right? Re-read only the lessons you missed — not the whole chapter.

A · MATHS — Algebra + Consecutive Numbers

Core idea

RRB algebra is mostly **identities** and **$x + 1/x$** type questions, plus simple linear and quadratic equations. Learn the identities so well that you recognise them instantly. For word problems with consecutive numbers, call the middle number **n** — the sum becomes $3n$ and the equation solves in one step.

Formulas

Concept	Formula / rule
Squares	$(a + b)^2 = a^2 + 2ab + b^2$ · $(a - b)^2 = a^2 - 2ab + b^2$ · $a^2 - b^2 = (a + b)(a - b)$
Cubes	$(a + b)^3 = a^3 + b^3 + 3ab(a + b)$ · $a^3 + b^3 = (a + b)(a^2 - ab + b^2)$ · $a^3 - b^3 = (a - b)(a^2 + ab + b^2)$
Three variables	$a^3 + b^3 + c^3 - 3abc = (a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca)$ · if $a + b + c = 0 \rightarrow a^3 + b^3 + c^3 = 3abc$
$x + 1/x = k$	$x^2 + 1/x^2 = k^2 - 2$ · $x^3 + 1/x^3 = k^3 - 3k$
$x - 1/x = k$	$x^2 + 1/x^2 = k^2 + 2$
Quadratic $ax^2 + bx + c = 0$	Sum of roots $-b/a$ · product c/a
Consecutive	Integers $n - 1, n, n + 1$ · odd or even: $n - 2, n, n + 2$

Patterns — the method and one solved example each

Pattern	Method	Solved example
1. $x + 1/x$	$k^2 - 2$	$x + 1/x = 5 \rightarrow x^2 + 1/x^2 = \mathbf{23}$
2. Cube version	$k^3 - 3k$	$x + 1/x = 3 \rightarrow x^3 + 1/x^3 = 27 - 9 = \mathbf{18}$
3. $a^2 + b^2$	$(a + b)^2 - 2ab$	$a + b = 10, ab = 21 \rightarrow 100 - 42 = \mathbf{58}$
4. $a + b + c = 0$	$3abc$	$(a^3 + b^3 + c^3)/abc = \mathbf{3}$
5. Difference of squares	$(a + b)(a - b)$	$101^2 - 99^2 = 200 \times 2 = \mathbf{400}$
6. Near 100	$a^2 - b^2$	$97 \times 103 = 100^2 - 3^2 = \mathbf{9991}$
7. Linear	Collect terms	$3x - 5 = 2x + 7 \rightarrow \mathbf{x = 12}$
8. Quadratic	Factorise	$x^2 - 5x + 6 = 0 \rightarrow (x - 2)(x - 3) \rightarrow \mathbf{2, 3}$ (sum 5, product 6)
9. Three consecutive	Middle = sum $\div 3$	Sum 72 $\rightarrow$ middle 24 $\rightarrow \mathbf{23, 24, 25}$
10. Consecutive odd	Middle = sum $\div 3$	Sum of three consecutive odd numbers 75 $\rightarrow \mathbf{23, 25, 27}$
11. Consecutive even product	Near $\sqrt{}$	Product 168 $\rightarrow \sqrt{168} \approx 13 \rightarrow \mathbf{12 \text{ and } 14}$
12. Sum and difference	Half-sum, half-difference	Sum 25, difference 7 $\rightarrow \mathbf{16 \text{ and } 9}$

Traps

- $x^3 + 1/x^3 = k^3 - \mathbf{3k}$, not $k^3 - 3$.
- $(a - b)^2$ has $-2ab$; $a^2 + b^2$ is never $(a + b)^2$.
- Consecutive odd/even numbers differ by **2**, not 1.

B · REASONING — Mixed Puzzles

Comparison puzzles (taller, heavier, more marks): write a chain with '>' signs, joining one clue at a time. **Distribution puzzles** (who likes what, who lives where): make a grid of people × options and use ✓ and ✗. A row with three ✗ out of four options tells you the fourth.

	Delhi	Mumbai	Pune	Goa
A	✗	✗	✗	✓
B	✗	✓	✗	✗
C	✗	✗	✓	✗
D	✓	✗	✗	✗

Grid above for: A lives in Goa · B doesn't live in Delhi or Pune · C doesn't live in Delhi.

Patterns, with solved examples

Pattern	Example	Solution
Comparison	A is taller than B but shorter than C. D is taller than C. E is the shortest. Tallest? Second tallest?	$D > C > A > B > E \rightarrow \mathbf{D, then C}$
Comparison	P scored more than Q; R less than Q; S more than P. Who scored least?	$S > P > Q > R \rightarrow \mathbf{R}$
Distribution	Amit, Bina, Chetan like tea, coffee, juice (one each). Amit likes neither tea nor juice. Bina doesn't like tea.	Amit coffee, Bina juice, Chetan tea
Grid	The city grid above	A Goa, B Mumbai, C Pune, D Delhi
Stack	5 boxes in a stack. S is at the top, R at the bottom, T immediately above R, P somewhere above Q. Order top to bottom?	S, P, Q, T, R

Speed rule

Use the grid even when the puzzle looks easy — the ✗ marks do the thinking for you.

C · SCIENCE — Biology 2: Food Production, Agriculture, Nutrition & Digestion

8 lessons. Food production and agriculture were 20 of 210 science questions in the 2026 papers (R) — the single biggest biology topic. Vitamins and deficiency diseases are classic one-mark facts.

Lesson 1 · Crop seasons and the revolutions (about 7 min)

Season	Sown / harvested	Crops
Kharif	With the monsoon, June–July → Sept–Oct	Paddy (rice), maize, soybean, groundnut, cotton, pigeon pea (arhar), green gram, black gram
Rabi	Winter, Oct–Nov → March–April	Wheat, gram, peas, mustard, linseed, barley
Zaid	Short summer season between them	Watermelon, muskmelon, cucumber, vegetables

Revolution	Product	Key name
Green	Food grains (high-yielding wheat, rice)	M. S. Swaminathan in India; Norman Borlaug worldwide
White (Operation Flood)	Milk	Verghese Kurien
Blue	Fish	—
Yellow	Oilseeds	—

Crop improvement: **hybridisation** (crossing varieties) and **genetic modification** — e.g. **Bt cotton**, made pest-resistant with a bacterial gene.

Quick check 1 — answer out loud before moving on (answers at the end)

a) Is wheat a kharif or a rabi crop? b) Who is called the father of the Green Revolution in India?

Lesson 2 · Plant nutrients, manure and fertilisers (about 8 min)

Plants need **16 nutrients**. Carbon, hydrogen and oxygen come from air and water; the other 13 from the soil. **Macronutrients**: N, P, K, Ca, Mg, S. **Micronutrients**: Fe, Mn, B, Zn, Cu, Mo, Cl.

	Manure	Fertiliser
Made from	Animal dung and plant waste (organic)	Chemicals in factories (urea, superphosphate, NPK)
Nutrients	Fewer, but adds humus and improves soil texture	Rich in specific nutrients (N, P, K)
Overuse	Safe	Harms soil microbes; run-off causes water pollution (eutrophication)

Vermicompost uses earthworms (red worms). **Green manure**: plants like sun hemp ploughed back into the soil. **Nitrogen fixation**: **Rhizobium** bacteria live in root nodules of **legumes** (peas, beans, gram) — so crop rotation with legumes refills nitrogen. Biofertilisers: Rhizobium, Azotobacter, blue-green algae (Anabaena) in paddy fields.

Quick check 2 — answer out loud before moving on (answers at the end)

a) Which bacterium lives in the root nodules of legumes? b) Name the organism used in vermicompost.

Lesson 3 · Cropping patterns, irrigation and protection (about 7 min)

Practice	Meaning
Mixed cropping	Two or more crops grown together on the same land (wheat + gram)
Intercropping	Crops in a definite row pattern (soybean + maize)
Crop rotation	Different crops on the same field in planned seasons — often a legume to restore nitrogen

Irrigation: wells, canals, river-lift, tanks. Water-saving modern methods: **sprinkler** and **drip** (drip is the most water-efficient — water falls drop by drop at the roots).

Weeds (Xanthium, Parthenium or 'congress grass') are controlled by weedicides such as **2,4-D**. Stored grain is spoiled by **biotic** factors (insects, rodents, fungi, bacteria) and **abiotic** ones (moisture, temperature); prevent by drying, cleaning and fumigation.

Quick check 3 — answer out loud before moving on (answers at the end)

a) Which irrigation method saves the most water? b) Name a common weedicide.

Lesson 4 · Animal husbandry, fisheries and bee-keeping (about 8 min)

Cattle: milch animals (milk) and draught animals (work). Indian breeds: Red Sindhi, Sahiwal; exotic: Jersey, Brown Swiss (long milk-producing period). **Poultry:** layers give eggs, broilers give meat; Aseel is an Indian breed.

Composite fish culture — fish	Feeding zone
Catla	Surface
Rohu	Middle zone
Mrigal, common carp	Bottom

Using all zones means no competition for food. Marine fish: pomfret, mackerel, tuna, sardine, Bombay duck.

Term	Rearing of
Apiculture	Honey bees (Indian bee <i>Apis cerana indica</i> ; Italian bee <i>Apis mellifera</i> — high yield)
Sericulture	Silkworms (silk) — fed on mulberry leaves
Pisciculture	Fish
Vermiculture	Earthworms

Quick check 4 — answer out loud before moving on (answers at the end)

a) Which fish feeds at the surface in composite fish culture? b) Rearing of silkworms is called?

Lesson 5 · Nutrients in food (about 7 min)

Nutrient	Job	Test / note
Carbohydrates	Main energy source (≈ 4 kcal/g)	Starch + iodine → blue-black
Fats	Concentrated energy (≈ 9 kcal/g)	Leaves an oily patch on paper
Proteins	Body-building and repair (≈ 4 kcal/g)	$\text{CuSO}_4 + \text{NaOH} \rightarrow$ violet
Vitamins, minerals	Protect from disease; needed in small amounts	See next two lessons
Roughage (fibre)	Helps bowel movement	No energy
Water	Transport, temperature control	$\approx 60\text{--}70\%$ of body weight

Protein deficiency in children: **kwashiorkor** (swollen belly). Protein + calorie deficiency: **marasmus** (very thin).

Quick check 5 — answer out loud before moving on (answers at the end)

a) Iodine solution turns blue-black with which nutrient? b) Disease caused by protein deficiency in children?

Lesson 6 · Vitamins (about 9 min)

Vitamin	Chemical name	Deficiency disease	Sources
A	Retinol	Night blindness , xerophthalmia	Carrot, milk, liver, papaya
B1	Thiamine	Beriberi	Whole grains, pulses
B2	Riboflavin	Cracked lips and skin	Milk, eggs
B3	Niacin	Pellagra	Meat, grains
B9	Folic acid	Anaemia	Green leafy vegetables
B12	Cobalamin (contains cobalt)	Pernicious anaemia	Animal foods
C	Ascorbic acid	Scurvy (bleeding gums)	Citrus, amla; destroyed by heating
D	Calciferol	Rickets (children), osteomalacia (adults)	Made in skin in sunlight
E	Tocopherol	Reproductive problems	Vegetable oils, nuts
K	Phylloquinone	Blood does not clot	Green vegetables

Fat-soluble: A, D, E, K (stored in the body). **Water-soluble: B and C**. The word 'vitamine' was coined by **Casimir Funk**.

Quick check 6 — answer out loud before moving on (answers at the end)

a) Which vitamin is made in the skin in sunlight? b) Lack of vitamin C causes? c) Which vitamin contains cobalt?

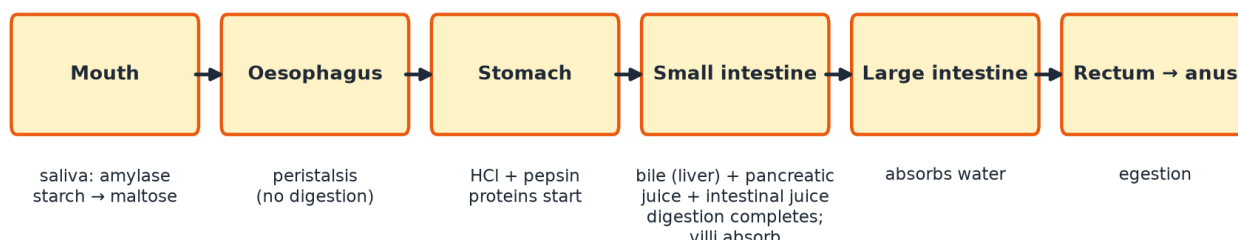
Lesson 7 · Minerals (about 5 min)

Mineral	Needed for	Deficiency
Calcium	Bones, teeth, clotting	Weak bones (osteoporosis)
Iron	Haemoglobin	Anaemia
Iodine	Thyroxine (thyroid hormone)	Goitre
Phosphorus	Bones, teeth, ATP	Weak bones
Sodium, potassium	Nerve impulses, fluid balance	Muscle cramps
Fluorine	Tooth enamel	Tooth decay (excess → fluorosis)
Magnesium	Chlorophyll in plants	Yellow leaves in plants

Quick check 7 — answer out loud before moving on (answers at the end)

a) Lack of iodine causes? b) Which mineral is part of haemoglobin?

Lesson 8 · Digestion (about 10 min)



Where	Juice / enzyme	Acts on
Mouth	Saliva — salivary amylase (ptyalin)	Starch → maltose
Stomach	HCl (kills germs, makes it acidic) · pepsin · mucus (protects the wall)	Proteins → peptides
Liver → gall bladder	Bile — no enzymes; alkaline	Emulsifies fats (breaks them into droplets)
Pancreas	Pancreatic juice: trypsin · amylase · lipase	Proteins · starch · fats
Small intestine	Intestinal juice — completes digestion	Food absorbed by villi
Large intestine	—	Absorbs water

The **liver** is the largest gland. The **pancreas** is both an exocrine (juice) and endocrine (insulin) gland. The small intestine is the longest part of the gut; herbivores have a longer one to digest cellulose.

Quick check 8 — answer out loud before moving on (answers at the end)

a) Which juice has no enzymes but emulsifies fats? b) Where does protein digestion begin?

Science 1-page summary — revise from this tomorrow

Topic	Must remember
Seasons	Kharif (monsoon): rice, maize, cotton, soybean, arhar · rabi (winter): wheat, gram, mustard, peas · zaid: melons
Revolutions	Green — grains, Swaminathan (India), Borlaug · White — milk, Kurien · Blue — fish · Yellow — oilseeds
Soil	16 nutrients · Rhizobium in legume nodules fixes N · vermicompost earthworms · manure adds humus · drip saves most water
Animals	Catla surface, Rohu middle, Mrigal bottom · apiculture bees · sericulture silk · Jersey, Brown Swiss exotic cows
Nutrients	Starch + iodine blue-black · protein test violet · kwashiorkor (protein), marasmus (protein + calories)
Vitamins	A night blindness · B1 beriberi · B3 pellagra · B12 cobalt · C scurvy · D rickets, sunlight · K clotting · fat-soluble A D E K
Minerals	Iron → haemoglobin, anaemia · iodine → thyroxine, goitre · calcium bones · fluorine enamel
Digestion	Saliva amylase starch · stomach HCl + pepsin protein · bile emulsifies fat (no enzyme) · pancreas trypsin, lipase · villi absorb

Science practice set — 20 questions · target 6 minutes

#	Question
1	Which is a rabi crop? (a) Wheat (b) Paddy (c) Maize (d) Cotton
2	Which is a kharif crop? (a) Wheat (b) Paddy (c) Mustard (d) Gram
3	The White Revolution is associated with: (a) fish (b) oilseeds (c) milk (d) food grains
4	Bt cotton is resistant to: (a) insect pests (b) drought (c) frost (d) salty soil
5	Rhizobium bacteria live in the root nodules of: (a) wheat (b) rice (c) maize (d) legumes
6	Vermicompost is made with the help of: (a) honey bees (b) silkworms (c) fungi only (d) earthworms
7	Which is a micronutrient for plants? (a) Nitrogen (b) Zinc (c) Potassium (d) Phosphorus
8	The most water-efficient irrigation method is: (a) canal irrigation (b) flood irrigation (c) well irrigation (d) drip irrigation
9	2,4-D is used as a: (a) fertiliser (b) insecticide (c) fungicide (d) weedicide
10	In composite fish culture, Catla feeds at the: (a) bottom (b) middle zone (c) surface (d) plant roots
11	Sericulture is the rearing of: (a) honey bees (b) fish (c) earthworms (d) silkworms
12	Starch gives a blue-black colour with: (a) copper sulphate (b) phenolphthalein (c) litmus (d) iodine
13	Kwashiorkor is caused by deficiency of: (a) protein (b) vitamin A (c) iron (d) iodine
14	Night blindness is caused by deficiency of vitamin: (a) A (b) B1 (c) C (d) D
15	Scurvy is caused by deficiency of vitamin: (a) A (b) C (c) D (d) K
16	The vitamin made in the skin in sunlight is vitamin: (a) A (b) D (c) C (d) K
17	Which vitamin helps blood to clot? (a) Vitamin A (b) Vitamin K (c) Vitamin C (d) Vitamin E
18	Goitre is caused by deficiency of: (a) iron (b) iodine (c) calcium (d) vitamin D
19	Bile is produced by the: (a) gall bladder (b) pancreas (c) liver (d) stomach
20	Pepsin acts on: (a) starch (b) fats (c) proteins (d) cellulose

Score	Time taken	Wrong Q numbers	Mistake type (concept / formula / calc / careless / recall)
/ 20			

Answers

Science practice set

#	Ans	Why / lesson to re-read
1	a	Lesson 1
2	b	Lesson 1
3	c	Lesson 1
4	a	Lesson 1
5	d	Lesson 2
6	d	Lesson 2
7	b	Lesson 2
8	d	Lesson 3
9	d	Lesson 3
10	c	Lesson 4
11	d	Lesson 4
12	d	Lesson 5
13	a	Lesson 5
14	a	Lesson 6
15	b	Lesson 6
16	b	Lesson 6
17	b	Lesson 6
18	b	Lesson 7
19	c	Lesson 8 — gall bladder only stores it
20	c	Lesson 8

Quick checks

QC	Answers
1	a) Rabi b) M. S. Swaminathan
2	a) Rhizobium b) Earthworms (red worms)
3	a) Drip irrigation b) 2,4-D
4	a) Catla b) Sericulture
5	a) Starch (carbohydrate) b) Kwashiorkor
6	a) Vitamin D b) Scurvy c) Vitamin B12
7	a) Goitre b) Iron
8	a) Bile b) In the stomach (pepsin)

Day 13 recall (Block 0)

#	Answer	#	Answer
1	Robert Hooke	6	Muscle to bone
2	Mitochondria	7	No — involuntary (but striated)
3	Lysosomes	8	R. H. Whittaker
4	46	9	38,808
5	Xylem	10	Clockwise

Send me tonight

Day 13 recall score /10 · science practice score /20 and time · Maths and Reasoning: attempted, correct, average seconds per question · the question numbers you got wrong.